



Linearity improvement of chirped distributed feedback laser diodes based on an analog electro-optical phase-locked loop

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Abstract: A linearity improvement method for frequency-modulated distributed feedback laser diodes (DFB-LD) is proposed and demonstrated based on a pre-distortion signal and an electro-optical phase-locked loop (EO-PLL). The pre-distortion signal is used to reduce the deterministic frequency errors. The EO-PLL is further used to suppress the stochastic frequency noise and enhance the coherence of the DFB-LD. In the EO-PLL, the DFB-LD output is transmitted through a Mach-Zehnder interferometer (MZI) and detected by a photodetector (PD) to get a beat note signal, which denotes the nonlinearity of the chirp. A mixing signal, achieved by mixing the beat note signal with a fixed frequency reference signal, is then filtered by a proportional integral filter (PIF) and feedback to the DFB-LD to reduce the stochastic frequency noise in the chirp. The EO-PLL bandwidth can be adjusted by tuning the PIF response. Consequently, a linear chirp optical signal with an enhanced linearity is generated from the DFB-LD. In the experiment, 788- and 321-time linearity improvements are implemented when the loop bandwidths are about 100 and 550 kHz, respectively. Correspondingly, residual frequency errors of 1.65 and 4.85 MHz at 52- and 450-THz/s chirp rates are obtained.

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1. Introduction

Linear frequency modulation (LFM) laser sources are widely used in various significant fields, such as frequency-modulated continuous wave light detection and ranging (FMCW LiDAR) [1–4], optical frequency domain reflectometry (OFDR) [5], optical coherence tomography (OCT) [6,7] and microwave photonics signal generation and processing [8–12]. For these applications, there is a growing need for LFM laser sources with wide frequency chirp range, large chirp rate, and high chirp linearity [1–12]. Semiconductor lasers (SCLs) are the recommended choice for LFM sources due to their compact size, lightweight, low cost, ease of integration, and simplicity of tuning. However, owing to the inherent nonlinearity in the semiconductor tuning process, precise linear frequency control typically requires additional techniques.

The two most popular chirp linearization techniques are driving signal pre-distortion [13,14] and electro-optical phase-locked-loop (EO-PLL) [15–21]. The driving signal pre-distortion method compensates for the inherent nonlinear characteristics of the laser by pre-introducing a correction in the driving signal [13,14]. However, this method cannot suppress broadband random frequency noise, meaning it cannot enhance the coherent properties of semiconductor lasers. A powerful technique to suppress the broadband random frequency noise of the LFM lasers is the EO-PLL, which has a structure similar to the conventional electrical phase-locked loop, with the key difference that the electro-optical voltage-controlled oscillator (VCO) part consists of an integrator, a continuously tunable laser, an unbalanced Mach-Zehnder interferometer (MZI), and a photodetector [15,16]. In recent decades, several EO-PLL configurations have been demonstrated

to improve the linearity of the LFM lasers [16–21]. For example, a digital EO-PLL, where the beat note signals are converted into digital signals for mixing, filtering, and processing, is used to enhance the stability of linearization system settings [16,17]. However, complicated digital logic is necessary for the implementation of digital EO-PLL. The analog EO-PLL is another simple technique to improve the linearity of the LFM lasers [18–21]. In conventional analog EO-PLL with the phase frequency detector (PFD), an acousto-optic frequency shifter (AOF) is usually used to increase the frequency of the beat note signal for sufficient loop bandwidth [18–20]. However, using the AOF increases the complexity and cost of the system. A more straightforward analog EO-PLL structure is using a mixer to replace the AOF and the PFD [21]. However, the mixer output contains multiple frequency components, which affects phase locking accuracy and system stability [21].

In this paper, we propose and experimentally demonstrate an LFM distributed feedback laser diode (DFB-LD) based on a mixer-based analog EO-PLL, where a proportional-integral filter (PIF) is introduced to reduce the effect of the multiple frequency components and control the loop bandwidth [21]. In addition, a pre-distortion signal is added to the EO-PLL to reduce the deterministic frequency errors. Compared to previous EO-PLLs [16–21], the analog circuit design in the proposed EO-PLL combining a mixer and a PIF avoids the need for complex and expensive hardware, such as field-programmable gate arrays (FPGA) [16,17] and AOFs [18–20]. In the experiment, 788-time linearity improvements and 550-kHz loop bandwidths were obtained.

2. Principle

The schematic diagram of the EO-PLL-based LFM DFB-LD is shown in Fig. 1. First, a pre-distortion signal is injected into the DFB-LD to reduce the deterministic frequency errors. Then, the EO-PLL is implemented to suppress the stochastic frequency noise and enhance the coherence of the DFB-LD. In the EO-PLL, ninety percent of the output power of the LFM DFB-LD is tapped out as the application optical path, and ten percent of the power is coupled into the EO-PLL by a 10/90 optocoupler (OC1). The coupled signal is then transmitted through an MZI consisting of two 50/50 OCs (OC2 and OC3) and two optical fibers with a time delay difference of τ . The MZI-transmitted signal is detected by a balanced photodetector (BPD). The detected beat note signal is mixed with a reference signal via an electrical mixer. The mixed signal is then filtered by a PIF, processed by an integrator, combined with the pre-distorted signal via an electrical adder, and injected into the DFB-LD.

By injecting a pre-distortion drive signal into the DFB-LD, the laser can generate a linearly chirped frequency-modulated optical signal, and the optical frequency in a single chirp period can be written as

$$\omega(t) = \frac{d\phi(t)}{dt} = \omega_0 + \gamma t + \delta\omega(t), \quad 0 < t < \frac{T}{2} \quad (1)$$

where $\phi(t)$ is the optical phase, ω_0 is the initial frequency, $\delta\omega(t)$ is the frequency error during the chirp, $\gamma = 2B/T$ is the chirp rate, and T is the period of the symmetric triangular-wave signal, B is the frequency chirp range. The chirp nonlinearity is measured by an unbalanced MZI with a time delay difference of τ . The output optical signal of the MZI passes through a 2×2 OC into a balanced photodetector (BPD). The phase difference between the signals from the two arms of the MZI is obtained at the BPD and is given by

$$\phi_b(t) = \phi(t) - \phi(t - \tau) \approx \frac{d\phi(t)}{dt} \tau = \omega(t) \tau \quad (2)$$

The phase difference $\phi_b(t)$ is proportional to the instantaneous optical frequency as long as the propagation delay τ is small enough [21]. So, the phase error $\delta\phi_b(t)$ of the beat note signal has

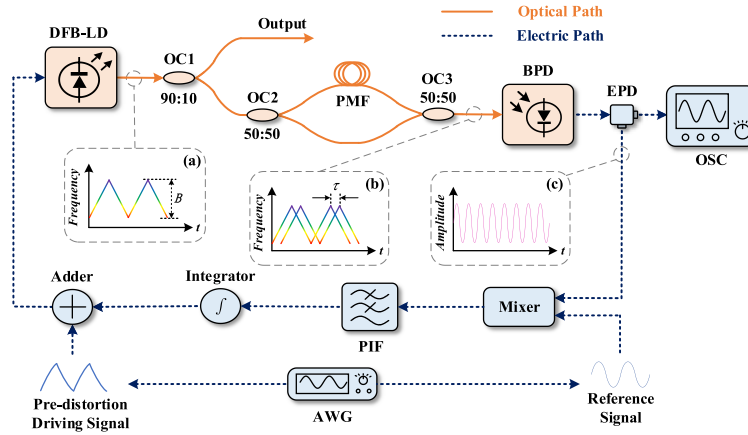


Fig. 1. Schematic diagram of the proposed linear frequency modulation laser source. (a) laser output signal in the time-frequency domain; (b) MZI output signal in the time-frequency domain; (c) BPD output signal in the time domain. DFB-LD: distributed feedback laser diodes; OC: optical coupler; PMF: polarization maintaining fiber; BPD: balanced photodetector; EPD: electrical power divider; OSC: oscilloscope; PIF: proportional integral filter; AWG: arbitrary waveform generator.

the following relationship with the frequency error of the chirp optical signal.

$$\delta\phi_b(t) = \delta\omega(t) \cdot \tau \quad (3)$$

As a result, the chirp nonlinearity can be measured by the phase of the beat note signal at the BPD [13]. Subsequently, the beat note signal and the reference signal with frequency ω_r are input to an analog mixer, and the signal from the mixer contains high harmonic combination frequency components in addition to the difference and sum frequency components. Undesired frequency components are filtered out by the PIF so that the final output contains only the difference frequency component, which is the intermediate frequency (IF) signal, and the signal voltage can be written as

$$V_M(t) = K_M[\cos(\phi_b(t) - \omega_r t)] \quad (4)$$

where K_M is the amplitude of the IF signal from the mixer. Setting the reference frequency $\omega_r = \gamma\tau$, the targeted down-conversion component contains the instantaneous frequency error given by

$$\theta_e(t) = \phi_b(t) - \omega_r t = \omega_0 \tau + \delta\omega(t) \cdot \tau \quad (5)$$

$\theta_e(t)$ serves as the feedback quantity of the loop, which includes the swept nonlinearity, phase noise of the laser such as phase random walk induced by the spontaneous emission, and the technical 1/f noise, assuming that laser intensity noise, photodiode shot noise, and electronic noise can be neglected. A larger delay τ induced from the MZI would improve the frequency discrimination, allowing the loop to detect and compensate for low-frequency phase error more accurately. However, an excessively large τ reduces the feedback speed, limits the achievable loop bandwidth, and reduces high-frequency noise suppression performance. While increasing the loop bandwidth can help mitigate this issue, excessive bandwidth may compromise the stability of the phase-locked loop. So, an optimal τ should be chosen to balance phase noise suppression performance and loop stability. The reference signal frequency is set to match that of the beat note signal, and the error signal generated by mixing the two signals is processed by the PIF,

whose transfer function is denoted as

$$F(s) = \frac{1 + \tau_2 s}{1 + \tau_1 s} = \frac{\tau_2}{\tau_1} \left(1 + \frac{1/\tau_2 - 1/\tau_1}{s + 1/\tau_1} \right) \quad (6)$$

where τ_1 and τ_2 are time constants with independent adjustability. The phase lead factor $1 + \tau_2 s$ introduces a phase lead correction effect at high frequencies, effectively enhancing loop stability. Equation (6) indicates that the transfer function can also be interpreted as a proportional-plus-imperfect integrator filter [22], where the proportional action is τ_2/τ_1 , and the integrator is instead of a low-pass filter with a low cutoff frequency. The proportional path provides the flexibility to determine the loop gain by adjusting the variable resistors, which directly affects the response speed and bandwidth of the PLL. The error signal is processed by the PIF to deliver a suitable control signal to the electro-optical VCO, denoted as $V_F(t)$. In the Laplace transform domain, the action of the filter is described by [22]

$$L\{V_F(t)\} = F(s)L\{V_M(t)\} \quad (7)$$

where $L\{\cdot\}$ indicates a Laplace transform. The PIF has the function of correcting phase errors at higher frequencies, optimizing the lock time and stability of the system, and playing a decisive role in adjusting the dynamic response and bandwidth of the loop. The phase difference stabilizes after a series of fluctuations, and the frequency error approaches zero. Ultimately, a linear chirp optical frequency output is achieved.

The frequency modulation response of the DFB-LD can be simplified as a low-pass filter [23], denoted as $H_{FM}(s)$. The transfer function of the electro-optical VCO $H_{VCO}(s)$ can be written as

$$H_{VCO}(s) = \frac{K_{INT} K_{SCL} \tau H_{FM}(s)}{s} \quad (8)$$

where K_{INT} is integrator gain, and K_{SCL} is semiconductor laser frequency modulation gain. The open-loop transfer function of the system can be written as

$$H_o(s) = \frac{K_M H_{VCO}(s) \cdot F(s)}{s} \cdot e^{-s\tau_d} \quad (9)$$

where τ_d is the sum of the optical path and circuit intrinsic time delays. The error transfer function of the system can be written as

$$H_e(s) = \frac{1}{1 + H_o(s)} \quad (10)$$

The 3-dB bandwidth of $H_e(s)$ can be selected to indicate the loop bandwidth. Compared to first-order phase-locked loop architectures lacking a loop filter, incorporating a PIF increases the flexibility to precisely control the loop bandwidth without relying solely on the gain selection of other fixed components. The added flexibility ensures robust operation across varying chirp rates while improving chirp linearity.

3. Experiment and result

An experiment is built according to the setup in Fig. 1 to demonstrate the proposed EO-PLL-based LFM DFB-LD. A 1550 nm commercial DFB-LD (CETC 1550-DFB) with a linewidth of about 500 kHz is used for the light source. The DFB-LD bias current is set to 210 mA with an output power of about 16 dBm. A triangular-wave signal generated by an arbitrary waveform generator (AWG) (Siglent SDG6052) is simultaneously injected into the DFB-LD through the laser driver, causing the frequency to chirp in a corresponding pattern. The MZI has a 1-m length difference

between its two arms, corresponding to a 5-ns time delay. Polarization maintaining fiber is used throughout the optical path. A BPD (Thorlabs PDB470C) converts the MZI outputs to electrical signals, and its output port is connected to an electrical power divider. One output of the divider is routed into the feedback loop. At the same time, the other is directed to a spectrum analyzer (Ceyear 4052 H) and an oscilloscope (Tektronix MSO54) for performance evaluation. The circuit portion of the loop consists of a mixer, a PIF, an integrator, and an adder, all fabricated on a small-size printed circuit board (PCB). The reference signal is generated from another channel of the AWG. The feedback and the pre-distortion signals are summed by the adder and injected into the DFB-LD to complete one calibration cycle.

Figure 2(a) shows the log-amplitude frequency response of the loop error transfer function $H_e(f)$ when the time constants are set as $\tau_1 = 96 \mu\text{s}$, $\tau_2 = 2 \mu\text{s}$. The DFB-LD has a frequency modulation gain of 0.375 GHz/mA, and its frequency modulation bandwidth is about 1 MHz. The modulation factor for the laser driver is 50 mA/V. In the case of $K_M = 0.2 \text{ V}$, $K_{INT} = 16.3 \text{ MHz}$, and $\tau_d = 20 \text{ ns}$, the total DC gain is 306.4 MHz, while the loop gain is 6.4 MHz [22]. The vertical dashed line indicates a loop bandwidth of approximately 550 kHz. The initial and pre-distortion triangular waves are shown in Figs. 2(b) and 2(c), respectively. A voltage signal with a frequency of 5 kHz and an amplitude of 2.4 V is converted into a current of 120 mA by the driver circuit and injected into the DFB-LD. The spectra of the beat note signals for the free-running, pre-distortion, and closed-loop conditions are presented in Fig. 2(d). It can be observed that under the free-running condition, the spectrum contains many undesired frequency components. After pre-distortion, excess frequency fluctuations are significantly reduced, and the frequency peak becomes visible. By setting the reference frequency to 2.25 MHz and achieving a closed-loop state, the frequency peak width of the beat note signal is further narrowed, and the signal-to-noise ratio is improved by 10 dB compared to the open-loop condition. It should be mentioned that we attempted to operate the feedback loop without the PIF. However, due to challenges in loop gain adjustment and multiple frequency components, the loop failed to lock, resulting in the spectra of the beat note signals nearly identical to the pre-distorted condition in Fig. 2(d). This instability underscores the PIF's crucial role in bandwidth control

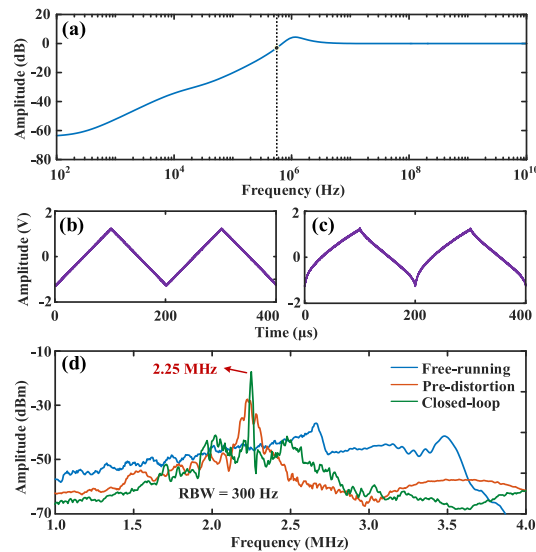


Fig. 2. (a) Frequency response of the loop error transfer function; (b) The initial and (c) pre-distortion triangular wave; (d) spectrum comparison of beat note signals.

and signal filtering, making it essential for linearity improvement and coherence enhancement in the EO-PLL.

To further evaluate the effectiveness of the proposed EO-PLL chirp linearization scheme, the time-domain signal acquired by the oscilloscope is subjected to short-time Fourier transform (STFT) [21], and the resulting time-frequency distribution is plotted in Fig. 3. In the free-running state, the frequency of the beat note signal fluctuates over a wide range. After pre-distortion, the frequency is stable except at the inflection point. When the loop is closed, the frequency stabilizes at a fixed value, confirming the results shown in Fig. 2(d). However, the frequency jitter persists at the inflection point, caused by transient loss of lock due to triangular wave modulation.

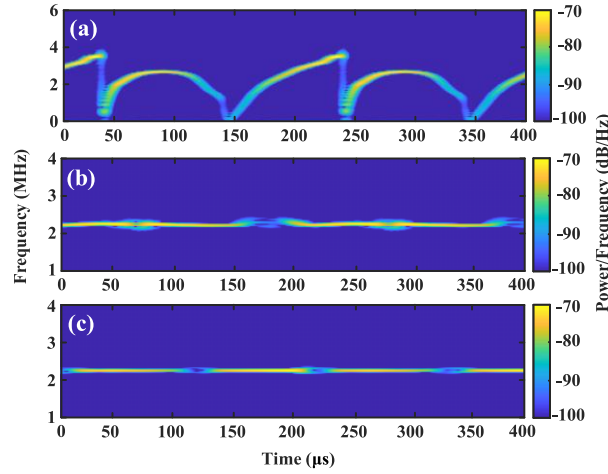


Fig. 3. The spectrogram of the beat note signals obtained using the STFT. (a) free-running state; (b) pre-distortion state; (c) closed-loop state.

To compare the chirp linearity before and after EO-PLL locking, the instantaneous frequency of the chirp is obtained by applying the Hilbert transform (HT) to the beat note signal [16,18,19] from the BPD, followed by a linear fit to the resulting frequency. Figures 4(a) and (b) show the positive and negative chirps in the free-running state, respectively. To mitigate the effect of nonlinearity at the inflection point, the central 80% of the region of interest (ROI) is used for calculation [13]. The frequency root-mean-square (RMS) errors are 1.47 GHz and 392.69 MHz, and the linear fit $1-r^2$ values are 1.65×10^{-2} and 1.07×10^{-3} , respectively. After pre-distortion, Figs. 4(c) and (d) show that the frequency RMS errors decrease to 33.53 and 17.23 MHz, and the linear fit $1-r^2$ values decrease to 1.07×10^{-5} and 2.87×10^{-6} , respectively. In comparison, in the closed-loop configuration, Figs. 4(e) and (f) illustrate the frequency RMS error of 4.58 and 4.68 MHz, and the linear fit $1-r^2$ of 2.04×10^{-7} and 2.14×10^{-7} for the positive and negative chirps, respectively. These data indicate that the proposed EO-PLL chirp linearization scheme optimizes linearity by a factor of 321 at a chirp rate of 450 THz/s.

To verify the performance of the EO-PLL at different chirp rates, the DFB-LD is modulated with a triangular wave at 500 Hz to produce a 52 GHz chirp range. The PIF parameters are adjusted to match the new chirp rate configuration, and the loop bandwidth is set to 100 kHz. Pre-distortion and EO-PLL control are applied in the same manner. With the loop closed, the reference frequency is set to 260 kHz, and the resulting beat note signal is analyzed. Figure 5 shows the optical frequency chirp curves generated by applying the HT to the beat note signals. In the free-running, pre-distorted, and closed-loop states, the frequency RMS errors for the positive chirps are 1.30 GHz, 16.35 MHz, and 1.65 MHz, while for the negative chirps, they are 241.35 MHz, 39.94 MHz, and 3.46 MHz, respectively. The linear fit $1-r^2$ values are

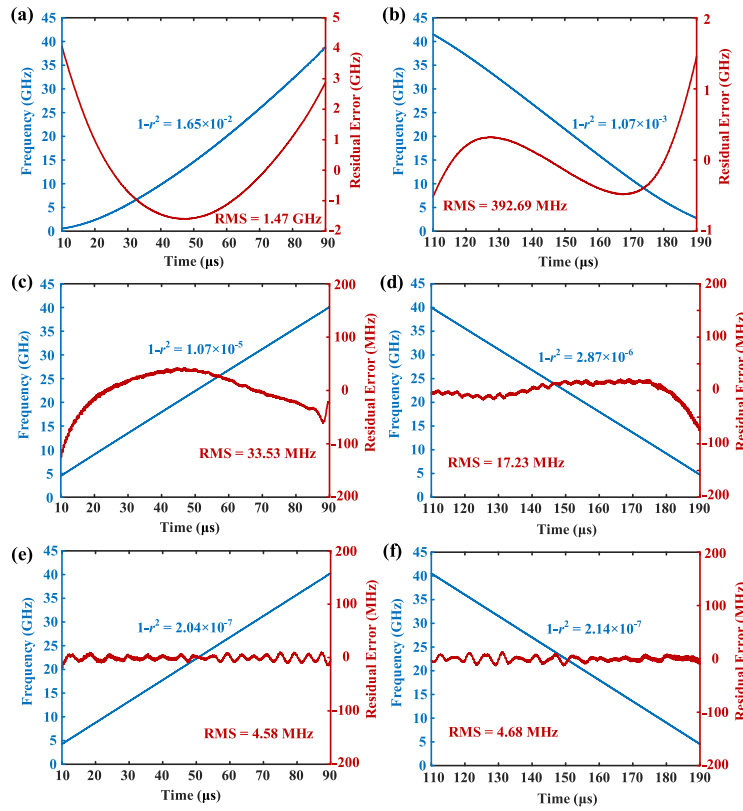


Fig. 4. Instantaneous frequencies and corresponding residuals for positive and negative chirps. (a)(b) free-running state; (c)(d) pre-distortion state; (e)(f) closed-loop state.

1.01×10^{-2} , 1.86×10^{-6} and 1.89×10^{-8} for the positive chirps, 3.36×10^{-4} , 1.12×10^{-5} , and 8.35×10^{-8} for the negative chirps, respectively. It is shown that the EO-PLL linearization scheme optimizes linearity by a factor of 788 at a chirp rate of 52 THz/s.

Table 1 compares our method with other analog EO-PLL-based laser chirp linearization schemes. It is worth mentioning that a PFD is generally used in conjunction with an AOF in the analog EO-PLLs of the comparison works [18–20]. One of the key reasons is that the sampling features of PFD-based EO-PLLs usually require a high reference frequency [24]. A mixer as a phase detector simplifies the system structure but compromises phase lock accuracy [21]. We introduce a PIF after the mixer, which enhances system stability, accuracy, and noise rejection while enabling bandwidth and dynamic response tuning to accommodate different chirp laser sources. The results demonstrate that high-frequency chirp linearity can be achieved with fast chirps using a simple analog EO-PLL structure. The proposed approach is adjustable for different chirp rates and has an advantage in linearity enhancement. The proposed approach achieves chirp rates up to 450 THz/s and provides flexible loop bandwidth tuning from 100 kHz to 550 kHz within a simplified analog architecture. These improvements make this approach suitable for high-speed applications, offering excellent linearity, simplicity, and flexibility.

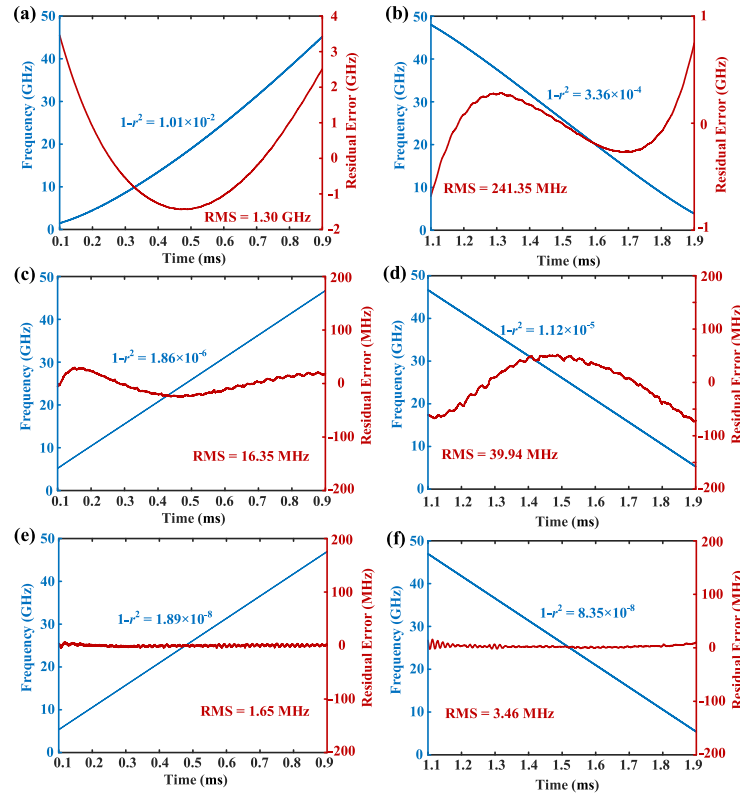


Fig. 5. Instantaneous frequencies and corresponding residuals for positive and negative chirps. (a)(b) free-running state; (c)(d) pre-distortion state; (e)(f) closed-loop state.

Table 1. Performance Comparison of analog EO-PLL-based laser FM linearization schemes

Structure	Loop Bandwidth (kHz)	Chirp range (GHz)	Chirp duration (ms)	Chirp rate (THz/s)	Frequency error (MHz)	Linearity enhancement factor
AOF + PFD + LP [18] ^a	-	4800	800	6	SD ^b 0.17	-
AOF + PFD + LP [19]	~1000	50	100	0.5	SD 0.089	-
AOF + PFD + LP [20]	1.9	8	50	0.16	-	-
Mixer [21]	±200	100	1	100	-	-
Mixer + PIF [This work]	550	36	0.08	450	RMS ^c 4.58	321
	100	41.6	0.8	52	RMS 1.65	788

^aAOF: acousto-optic frequency shifter; PFD: phase frequency detector; LP: loop filter.

^bSD: standard deviation.

^cRMS: root mean square.

4. Conclusions

In summary, an LFM DFB-LD with a linearity improvement based on a mixer-based analog EO-PLL was proposed and experimentally demonstrated. A proportional-integral filter (PIF) was introduced to reduce the effect of the multiple frequency components and control the loop bandwidth. A pre-distortion signal was added to the EO-PLL to reduce the deterministic frequency errors. 788- and 321-time linearity improvements were implemented when the loop bandwidths were about 100 and 550 kHz, respectively. Correspondingly, residual frequency errors of 1.65 and 4.85 MHz were obtained at 52- and 450-THz/s chirp rates. In particular, the uselessness of complicated digital logic and complex and expensive hardware such as FPGAs and AOFs within the feedback loop simplifies the system complexity, providing a high-performance, low-cost, and simple-structure LFM optical source solution for applications of FMCW LiDAR, OFDR, OCT, and MWP signal generation and processing.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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